

Applied Physics (NS1001)

Course Instructor(s):

Mrs.Aishaljaz, Dr.Mehwish Hassan, Dr.TashfeenZehra,
Mr.Kashif Ali

Section(s): (BS-SE, BS-CY, BS-AI, all sections)

Solution

Sessional-II Exam

Total Time (Hrs): 1

Total Marks: 40

Total Questions: 2

Date: Apr 13, 2026

Roll No

Course Section

Student Signature

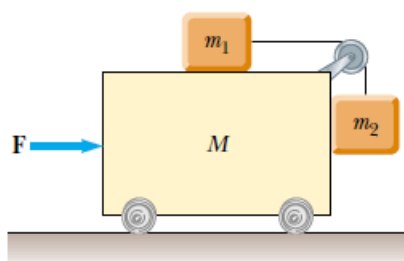
Do not write below this line.

Attempt all the questions.

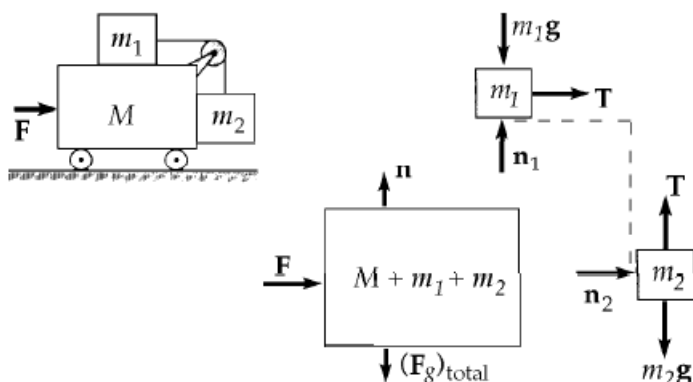
[CLO 1: Use vector algebra from Newtonian Mechanics to understand computer graphics/programming/simulations]

Question 1 [10]

What horizontal force must be applied to the cart shown in Figure, in order that the blocks remain stationary relative to the cart? Assume all surfaces, wheels, and pulley are frictionless. (Hint: Note that the force exerted by the string accelerates m_1 .) Draw Free body diagram also.



SOLUTION



National University of Computer and Emerging Sciences

Islamabad Campus

A cart of mass M is pushed to the right by a force F .

A block m_1 rests on top of the cart and is connected by a light string over a pulley to a block m_2 , which hangs along the side of the cart.

We want the force F such that:

- m_2 does not move up or down
- the whole arrangement moves together horizontally with acceleration a

Since m_2 does not move vertically, its vertical acceleration is zero.

So for block m_2 , vertical forces balance:

- Upward tension = T
- Downward weight = m_2g

Thus,

$$T - m_2g = 0$$

so,

$$T = m_2g$$

National University of Computer and Emerging Sciences
Islamabad Campus

Block m_1 moves horizontally with the cart.

The only horizontal force on m_1 is the string tension T .

So,

$$T = m_1 a$$

Now substitute $T = m_2 g$:

$$m_1 a = m_2 g$$

Hence,

$$a = \frac{m_2 g}{m_1}$$

Now consider the entire system:

$$M + m_1 + m_2$$

All of them move horizontally with the same acceleration a .

The only external horizontal force on this system is F . Therefore,

$$F = (M + m_1 + m_2) a$$

Substitute the value of a :

$$F = (M + m_1 + m_2) \left(\frac{m_2 g}{m_1} \right)$$

So the required force is

$$F = \frac{(M + m_1 + m_2) m_2 g}{m_1}$$

National University of Computer and Emerging Sciences
Islamabad Campus

[CLO 2: Use concepts of wave motion to define different oscillations graphically & mathematically]

Question 2 [10+5+10+5]

(i). The position of a 50 g oscillating mass is given by $x(t) = (2.0 \text{ cm})\cos(10t - \pi/4)$, where t is in s. Determine: (a). The amplitude (b). The period (c). The spring constant (d). The phase constant (e). The initial conditions $x(0)$ and $v(0)$ (f). The maximum speed (g). The total energy (h).

The velocity at $t = 0.40 \text{ s}$.

Solution

(a) Amplitude

$$A = 2.0 \text{ cm} = 0.020 \text{ m}$$

(b) Period

$$T = \frac{2\pi}{\omega} = \frac{2\pi}{10} = 0.628 \text{ s}$$

$$T = 0.628 \text{ s}$$

(c) Spring constant

For a mass-spring system,

$$\omega = \sqrt{\frac{k}{m}}$$

so

$$k = m\omega^2 = (0.050)(10^2) = 5.0 \text{ N/m}$$

$$k = 5.0 \text{ N/m}$$

(d) Phase constant

From the equation,

$$x(t) = A \cos(10t - \pi/4)$$

the phase constant is

$$\phi = \frac{\pi}{4} \text{ rad}$$

(e) Initial conditions $x(0)$ and $v(0)$

Position at $t = 0$:

$$x(0) = 0.020 \cos\left(-\frac{\pi}{4}\right)$$

$$x(0) = 0.020 \cdot \frac{\sqrt{2}}{2} = 0.0141 \text{ m}$$

$$x(0) = 1.41 \times 10^{-2} \text{ m} = 1.41 \text{ cm}$$

Velocity:

$$v(t) = \frac{dx}{dt} = -A\omega \sin(\omega t - \phi)$$

$$v(t) = -(0.020)(10) \sin(10t - \pi/4)$$

$$v(t) = -0.20 \sin(10t - \pi/4)$$

At $t = 0$:

$$v(0) = -0.20 \sin\left(-\frac{\pi}{4}\right)$$

$$v(0) = -0.20 \left(-\frac{\sqrt{2}}{2}\right) = 0.141 \text{ m/s}$$

$$v(0) = 0.141 \text{ m/s}$$

(f) Maximum speed

Maximum speed in SHM is

$$v_{\max} = A\omega$$

$$v_{\max} = (0.020)(10) = 0.20 \text{ m/s}$$

$$v_{\max} = 0.20 \text{ m/s}$$

(g) Total energy

$$E = \frac{1}{2}kA^2$$

$$E = \frac{1}{2}(5.0)(0.020)^2$$

$$E = 2.5(0.0004) = 0.0010 \text{ J}$$

$$E = 1.0 \times 10^{-3} \text{ J}$$

(h) Velocity at $t = 0.40 \text{ s}$

Use

$$v(t) = -0.20 \sin(10t - \pi/4)$$

At $t = 0.40$,

$$v(0.40) = -0.20 \sin(4 - \pi/4)$$

Since

$$4 - \frac{\pi}{4} \approx 3.2146$$

and

$$\sin(3.2146) \approx -0.0730$$

then

$$v(0.40) = -0.20(-0.0730) = 0.0146 \text{ m/s}$$

$$v(0.40 \text{ s}) \approx 1.46 \times 10^{-2} \text{ m/s}$$

(ii). Suppose that a simple pendulum consists of a small 60.0 g bob at the end of a cord of negligible mass. If the angle θ between the cord and the vertical is given by

$$\theta = (0.0800 \text{ rad})\cos [(4.43 \text{ rad/s})t + \phi],$$

National University of Computer and Emerging Sciences
Islamabad Campus

what are (a). the pendulum's length and (b). its maximum kinetic energy?

Solution : (a)

$$\omega = 4.43 \text{ rad/s}$$

Since $\omega = \sqrt{\frac{g}{L}}$, we solve for the length:

$$L = 0.499 \text{ m}$$

(b)

Using $v_m = \omega x_m = \omega L \theta_m$:

$$v_m = (4.43 \text{ rad/s})(0.499 \text{ m})(0.0800 \text{ rad})$$

With $m = 0.0600 \text{ kg}$, the maximum kinetic energy is:

$$\frac{1}{2}mv_m^2 = 9.40 \times 10^{-4} \text{ J}$$

(iii). A 50g hardboiled egg moves on the end of a spring with force constant $k = 25 \text{ N/m}$. Its initial displacement is 0.3 m. A damping force $F_x = -bV_x$ acts on the egg, and the amplitude of the motion decreases to 0.1m in 5.s. Calculate the magnitude of the damping constant b .

SOLUTION

$$\frac{A}{A_0} = e^{-\frac{b}{2m}t}$$

$$\frac{0.10}{0.30} = e^{-\frac{b}{2(0.050)} \cdot 5}$$

$$\frac{1}{3} = e^{-\frac{b}{0.1} \cdot 5}$$

$$\frac{1}{3} = e^{-50b}$$

$$\ln\left(\frac{1}{3}\right) = -50b$$

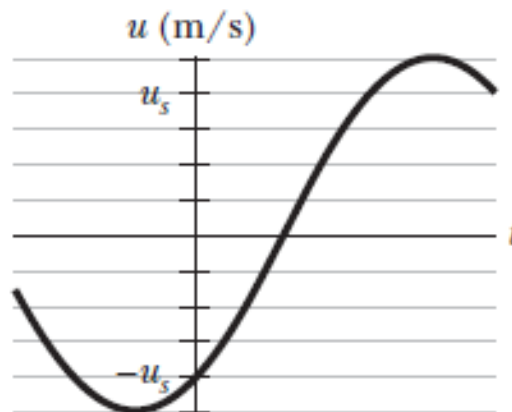
$$-1.0986 = -50b$$

$$b = \frac{1.0986}{50}$$

$$b \approx 0.022 \text{ kg/s}$$

$$b \approx 0.022 \text{ kg/s} = 22 \text{ g/s}$$

(iv). Figure shows the transverse velocity u versus time t of the point on a string at $x = 0$, as a wave passes through it. The scale on the vertical axis is set by $u_s = 4.0 \text{ m/s}$. The wave has the generic form $y(x, t) = y_m \sin(kx - \omega t + \phi)$. What then is ϕ ?



SOLUTION

$$u = -\omega y_m \cos(kx - \omega t + \phi)$$

Setting $x = 0$:

$$u = -\omega y_m \cos(-\omega t + \phi)$$

Differentiating with respect to time:

$$\frac{du}{dt} = -y_m \omega^2 \sin(-\omega t + \phi)$$

At $t = 0$:

$$\frac{du}{dt} = -y_m \omega^2 \sin \phi$$

Given the slope is positive:

$$-y_m \omega^2 \sin \phi > 0 \Rightarrow \sin \phi < 0$$

Thus, ϕ lies in the third or fourth quadrant.

Given:

$$u(0) = -4 \text{ m/s}, \quad u_{\max} = 5 \text{ m/s}$$

Also,

$$u_{\max} = \omega y_m$$

At $t = 0$:

$$-4 = -\omega y_m \cos \phi = -5 \cos \phi$$

$$\cos \phi = \frac{4}{5}$$

$$\phi = \cos^{-1}\left(\frac{4}{5}\right) = 0.6435 \text{ rad}$$

Since $\sin \phi < 0$, ϕ is in the fourth quadrant:

$$\phi = -0.6435 \text{ rad}$$

Formula Sheet

Simple Harmonic Oscillator (SHO) Equations

For Ideal SHO

1. Position:
 $x = x_m \cos (\omega t + \phi)$
2. Velocity:
 $v = -\omega x_m \sin (\omega t + \phi)$
3. Acceleration:
 $a = -\omega^2 x_m \cos (\omega t + \phi)$

National University of Computer and Emerging Sciences

Islamabad Campus

4. Angular Frequency:

$$\omega = \frac{2\pi}{T} = 2\pi f$$

$$\omega = \sqrt{\frac{k}{m}} \text{ (angular frequency)}$$

5. Period:

$$T = 2\pi \sqrt{\frac{m}{k}} \text{ (period)}$$

For Damped SHO

6. Position with damping:

$$x(t) = x_m e^{(-bt/2m)} \cos(\omega' t + \phi)$$

7. Modified Angular Frequency:

$$\omega' = \sqrt{\frac{k}{m} - \frac{b^2}{4m^2}}$$

8. Energy over time:

$$E(t) \approx \frac{1}{2} k x_m^2 e^{(-bt/m)}$$

Or

$$E(t) = E_0 e^{(-t/\tau)} \text{ where } \tau = m/b$$

Waves and Interference

1. Sinusoidal Function.

$$y(x, t) = y_m \sin(kx - \omega t)$$

2. Sinusoidal function with phase constant

$$y(x, t) = y_m \sin(kx - \omega t + \Phi)$$

3. angular wave number

$$k = \frac{2\pi}{\lambda}$$

4. Angular frequency, frequency and Time period relation

$$\frac{\omega}{2\pi} = f = \frac{1}{T}$$

5. Wave speed

$$v = \frac{\omega}{k} = \frac{\lambda}{T} = \lambda f$$

.....The End.....